LESSON PLAN · SCRATCH JUNIOR (K–2) BEGINNER: LITTLE CODERS

Lesson 10: Bug Hunt 🐞

Module 4 | 30 minutes | Debugging | KG–Grade 2 · ages 5–7

Objective: Find and fix a mistake (a bug) in a script by watching what the character does.

Success criteria

- I can watch a script and see where it goes wrong.
- I can fix one block.
- I can test again.

Key vocabulary

- **bug**: A mistake in code.
- **debug**: Find and fix a bug.
- **test**: Run it to check.

You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)

Timed lesson plan (30 min)

Time	Stage	What happens
5 min	**Warm-up (unplugged)**	Silly robot (5 min): the teacher follows a child's directions with one deliberate mistake; children find it.
5 min	**Watch**	Play the lesson video with captions; pause at each heading and ask a question.
8 min	**Build along**	Squash the bug: learners follow the steps.
5 min	**Perform**	Wrong way!, Too far!, Spot the bug
3 min	**Probe**	Missing step, I fixed it!
4 min	**Wrap-up**	Exit ticket: What do you do after you fix a block?

Unplugged warm-up

Silly robot (5 min): the teacher follows a child's directions with one deliberate mistake; children find it.

Teaching the idea

Debugging: observe behaviour, locate where it diverges from the goal, change one block, re-test. Bugs can be wrong direction, wrong number or missing blocks.

Analogy: A detective looking for clues.

Live demo: Run a buggy script and pause at the mistake; ask “which block did that?”.

Questions to ask

Ask	Listen for
Where did it go wrong?	Point to the moment.
Which block was running?	Find it.
How many blocks do you change at once?	One.

Differentiation

Support
Children point to each block as it runs (slow-motion).

Stretch
Make a buggy script for a partner to fix.

Common misconceptions

Misconception	How to fix it
A bug means start again	Fix one block.
Bugs mean I’m bad at coding	Everyone makes bugs!

Assessment

Use the Code Lab results (auto-checked), the quick checks and the Probe answers. Written answers arrive in your review queue.

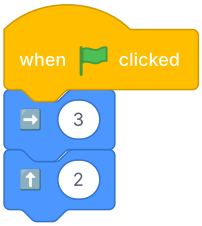
Exit ticket
What do you do after you fix a block?

Homework
Give someone wrong directions on purpose, and see if they can find the mistake.

Answer key — Kitabash activities

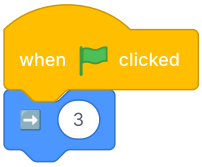
1. Wrong way! (perform · block code)

One solution:



2. Too far! (perform · block code)

One solution:

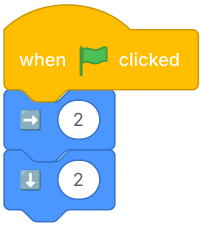


3. Spot the bug (perform · multiple choice)

Answer: A ⬇️ block should be ⬆️

4. Missing step (probe · block code)

One solution:



5. I fixed it! (probe · speaking)

Answer key — worksheet

1. **Key words:** bug = A mistake in code. · debug = Find and fix a bug. · test = Run it to check.
2. **Spot the bug:** A. A ⬇️ block should be ⬆️ — The arrow points the wrong way.
3. **Find the bug:** The ⬇️ should be ⬆️.

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.